

A Lower 1-Estimate Subcase of the Quantitative DN-S Question

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Status and source

This packet records a candidate partial result, likely valid, for Question 1 of Gonzalez and Martinon, *A quantitative approach to disjointly non-singular operators*, arXiv:2108.01052 [1]. The question appears in Section 7.1 on PDF page 11.

7.1. Open Questions. We finish the paper stating some open questions.

Question 1. *Is $\kappa_d \leq D \cdot \beta$ for some constant $D > 0$?*

The notation in the published PDF is transcribed as follows:

Question 1. Is $\kappa_d \leq D \cdot \beta$ for some constant $D > 0$?

The general question remains open in this packet. We prove it for Banach lattices with a uniform lower 1-estimate, including every abstract L_1 -space (AL-space), with the natural constant.

Definitions

Let E be a Banach lattice admitting an infinite disjoint sequence, let Y be a Banach space, and $T \in \mathcal{L}(E, Y)$. Write $d(E)$ for the set of disjoint sequences of nonzero vectors in E , and let $j(S) = \inf_{\|x\|=1} \|Sx\|$ be the injection modulus. The two quantities used below are

$$\tau_d(T) = \sup_{(x_n) \in d(E)} j(T|_{[x_n]}), \quad \kappa_d(T) = \inf_{(x_n) \in d(E)} \tau_d(T|_{[x_n]}),$$

and

$$\beta(T) = \inf \left\{ \liminf_{n \rightarrow \infty} \|Tx_n\| : (x_n) \subset S_E \text{ is disjoint} \right\}.$$

We say that E has a lower 1-estimate with constant $C \geq 1$ if every finite disjoint positive family $z_1, \dots, z_m$ satisfies

$$\sum_{i=1}^m \|z_i\| \leq C \left\| \sum_{i=1}^m z_i \right\|.$$

In an AL-space the norm is additive on disjoint positive vectors, so one may take $C = 1$.

Proof Intuition

The definition of $\beta(T)$ supplies a normalized disjoint sequence whose individual image norms are all as close to $\beta(T)$ as desired after passing to a subsequence. A pointwise bound on basis vectors normally does not control an operator on their whole span. The lower 1-estimate is the missing mechanism: it makes every normalized disjoint sequence dominate the unit vector basis of ℓ_1 . The triangle inequality then upgrades the pointwise image bound to an operator-norm bound on the entire disjoint span. Finally, κ_d is no larger than the τ_d -quantity of this one span, and that quantity is no larger than the restriction norm.

The partial result

Theorem 1. *Suppose that E admits an infinite disjoint sequence and has a lower 1-estimate with constant $C \geq 1$. Then for every Banach space Y and every $T \in \mathcal{L}(E, Y)$,*

$$\kappa_d(T) \leq C \beta(T).$$

No order-continuity hypothesis is needed for this estimate.

Proof. Fix $\varepsilon > 0$. By the definition of $\beta(T)$, there is a normalized disjoint sequence (u_n) such that

$$\liminf_{n \rightarrow \infty} \|Tu_n\| < \beta(T) + \varepsilon.$$

Passing to a subsequence and relabeling, we may assume that

$$\|Tu_n\| \leq \beta(T) + 2\varepsilon \quad (n \in \mathbb{N}).$$

For finitely supported scalars (a_n) , disjointness gives the lattice identity

$$\left| \sum_n a_n u_n \right| = \sum_n |a_n| |u_n|.$$

Since the u_n are normalized, the lower 1-estimate therefore yields

$$\sum_n |a_n| \leq C \left\| \sum_n a_n u_n \right\|.$$

Consequently,

$$\begin{aligned} \left\| T \left(\sum_n a_n u_n \right) \right\| &\leq \sum_n |a_n| \|Tu_n\| \\ &\leq C(\beta(T) + 2\varepsilon) \left\| \sum_n a_n u_n \right\|. \end{aligned}$$

By density, this proves

$$\|T|_{[u_n]}\| \leq C(\beta(T) + 2\varepsilon).$$

Every injection modulus occurring in $\tau_d(T|_{[u_n]})$ is bounded above by this restriction norm. Hence

$$\kappa_d(T) \leq \tau_d(T|_{[u_n]}) \leq \|T|_{[u_n]}\| \leq C(\beta(T) + 2\varepsilon).$$

Letting $\varepsilon \downarrow 0$ proves the theorem. □

Corollary 1. *If E is an AL-space, then every $T \in \mathcal{L}(E, Y)$ satisfies*

$$\kappa_d(T) \leq \beta(T).$$

Thus Question 1 has an affirmative answer with $D = 1$ on every abstract L_1 -space. The constant is optimal whenever E contains an infinite normalized disjoint sequence.

Proof. An AL-space has the lower 1-estimate with $C = 1$, so the inequality is Theorem 1. For sharpness, take $T = I_E$. Then every normalized disjoint sequence satisfies $\|Ix_n\| = 1$, so $\beta(I_E) = 1$. Also every restriction of the identity has injection modulus 1, whence $\kappa_d(I_E) = 1$. □

Verification notes

The proof is purely analytic and uses no computation. The potentially delicate step is the scalar-coefficient inequality. It follows directly from the lattice modulus identity for disjoint vectors, so it is valid over both the real and complex scalars. The argument does not assume that T is DN-S and remains valid when $\beta(T) = 0$.

An adversarial step check is included as `verification_report.md`. Its verdict is *likely valid*; no missing case or external theorem dependency was found.

Limitations and novelty check

This is a partial result only. It supplies no universal constant for order continuous lattices without a lower 1-estimate, and it does not settle the source question for L_p when $p > 1$.

Before promotion, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2108.01052, the exact κ_d - β comparison, and lower 1-estimate/AL-space variants. Related packets on dispersed quotient maps and LSS/DSS operators were inspected, but no duplicate of this theorem was found. A bounded web search for the exact question, the paper title, and the notation found the source paper and adjacent later DN-S papers, but no explicit later answer to this subcase. This is not a literature-complete novelty claim.

Human review recommendation

Send to a Banach-lattice expert as a candidate partial result. The reviewer should verify that the published $\kappa_d = i_d\tau_d$ convention is matched exactly and confirm that no broader known lower-estimate theorem already subsumes the observation.

References

- [1] M. Gonzalez and A. Martinon, *A quantitative approach to disjointly non-singular operators*, Rev. R. Acad. Cienc. Exactas Fis. Nat. Ser. A Mat. RACSAM **115** (2021), Article 185; arXiv:2108.01052; DOI: 10.1007/s13398-021-01124-1.